

Hematologic Medicine in the Athlete

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Athletic performance depends on proper functioning of the blood. From problems with the oxygen-carrying function of red blood cells to the prevention of bleeding by the hemostatic system, many hematologic issues can affect athletes adversely. These hematologic issues include both acquired and inherited disorders that can affect athletes of all ages. This chapter reviews many of the hematologic issues that may arise in caring for athletes, focusing on those relating to red blood cells and the hemostatic system.

DISORDERS OF RED BLOOD CELLS

Definition

Anemia is defined as a reduction of red blood cells below the normal range, with the normal range differing for men and women. Hemoglobin and hematocrit are commonly used to identify the red blood cell concentration in the blood. Mild anemia can be asymptomatic, although compared with more sedentary persons, athletes generally notice it much earlier because of its effects on their performance. It is important to identify the underlying cause of the anemia, which can be due to decreased red cell production or increased red cell destruction.

Hemoglobin is the oxygen-carrying protein in red blood cells. Normal hemoglobin (hemoglobin A) consists of two alpha chains and two beta chains.¹ Inherited disorders of hemoglobin are referred to as *hemoglobinopathies* or *thalassemia*. A number of hemoglobinopathies have been identified that result from mutations in the alpha or beta chains and have varying degrees of anemia and symptoms. The most common hemoglobin mutation in the United States is hemoglobin S, which causes sickle cell disease in persons with two copies of the mutated gene. *Thalassemia* refers to decreased production of normal alpha or beta chains and can be clinically silent or markedly symptomatic.² Thalassemias are referred to as alpha or beta, depending on which chain is affected. Thalassemias increase in clinical severity as the number of genes affected increases. In general, persons with one or two mutations are asymptomatic, a condition referred to as *thalassemia minor*.

It also appears that some athletes have anemia that is caused by mechanisms not seen in nonathletes, including dilutional “pseudoanemia” and exercise-related intravascular hemolysis. Pseudoanemia refers to a temporary condition that occurs as a result of training-related expansion of the plasma volume.³ The degree of volume expansion relates to the duration and intensity of exercise and can result in a dilutional drop in the hemoglobin

concentration.⁴ Intravascular hemolysis is also common in athletes. Initially referred to as *march hemoglobinuria* or *foot-strike hemoglobinuria*, the mechanism for hemolysis was thought to be induced by mechanical damage to red cells with each foot strike.⁵ However, others suggest that the effects of contracting muscles on red cells may contribute to hemolysis and the development of anemia in athletes.⁶

Epidemiology

Anemia is a condition commonly experienced by persons in the United States. Rates are highest in children and adult women, largely because of iron deficiency. The most recent National Health and Nutrition Examination Surveys (2003–2012) estimate that 3.5% of men and 7.6% of women in the United States are anemic, with the highest rates among menstruating women.⁷ Iron deficiency is the most common cause of anemia in the United States and is the most common nutritional deficiency worldwide.⁸ A recent retrospective study evaluated the results of iron-related testing in 2749 individuals at a National Collegiate Athletic Association (NCAA) Division 1 institution.⁹ In women, 2.2% had iron deficiency anemia and another 30.9% had iron deficiency without anemia. The incidence for men was much lower, 1.2% with iron deficiency anemia and 2.9% with iron deficiency without anemia.

Hemoglobin S is the most common inherited blood disorder in the United States. It is estimated that 1 in 12 African Americans carries one copy of the mutated gene, a condition referred to as *sickle cell trait*.¹⁰ The thalassemias are also common and are considered the most prevalent genetic mutation worldwide.² Incidence varies geographically; alpha thalassemia is frequent in Southeast Asia and western Africa, whereas beta thalassemia is more common in the Mediterranean region of Europe.

Pathophysiology

As previously stated, anemia is caused by either decreased production or increased destruction of red blood cells. Red blood cell production problems (Table 16.1) can be due to a vitamin or mineral deficiency, inflammation, erythropoietin deficiency, or a primary bone marrow disorder. Iron deficiency is the most common cause and is most often the result of chronic blood loss. Menstruation or chronic gastrointestinal blood loss will lead to iron deficiency if oral absorption is not able to balance iron loss. Other avenues for iron loss in athletes are hemolysis leading to hemoglobinuria and increased sweat production.¹⁰

Abstract

Athletic performance depends on proper functioning of the blood. From problems with the oxygen-carrying function of red blood cells to the prevention of bleeding by the hemostatic system, many hematologic issues can adversely affect athletes. These hematologic issues include both acquired and inherited disorders that can affect athletes of all ages. This chapter reviews many of the hematologic issues that may arise while caring for athletes, focusing on those relating to red blood cells and the hemostatic system.

Keywords

anemia
coagulation
iron deficiency
sickle cell trait
thrombocytopenia

TABLE 16.1 Types of Anemia in Athletes and Treatment Recommendations

Condition	Frequency	Hemoglobin	Ferritin	Treatment Required
Dilutional pseudoanemia	Common	Normal to mild decrease	Normal	No
Iron deficiency without anemia	Common	Normal	Decreased	Controversial
Iron deficiency anemia	Less common	Decreased	Decreased	Iron replacement indicated
Thalassemia minor	Less common	Normal to mild decrease	Normal to elevated	No

Another potential cause of iron deficiency in athletes is celiac disease, which results in iron malabsorption and can occur even in the absence of the gastrointestinal symptoms that are suggestive of celiac disease.¹¹

Iron deficiency occurs in three stages.⁸ Stage I, also known as prelatent iron deficiency, is associated with an isolated decrease in serum ferritin. At this stage, stainable iron is not present in the bone marrow, but hemoglobin levels remain normal. Stage II is latent iron deficiency; at this stage, the ferritin drops further, serum iron and transferrin saturation decrease, and total iron-binding capacity rises. As in stage I, hemoglobin levels remain normal during stage II, although the mean corpuscular volume (MCV) of the red cells may start to decrease. However, in stage III, with progressive depletion of iron stores, an overt microcytic and hypochromic iron-deficiency anemia develops.

Inflammatory conditions have long been known to lead to decreased red cell production, but only recently has the pathophysiology been understood fully. Hepcidin, a protein produced by the liver in response to infection or inflammation, inhibits iron absorption and its release from storage sites for use by developing red blood cell precursors.¹² Although significant data are lacking in athletes, it appears that chronic inflammation from training can result in elevated hepcidin levels, contributing to disordered iron metabolism and resultant anemia.

Erythropoietin is an important hormone that promotes red blood cell production.¹ It is produced by the kidney but is also manufactured as a pharmaceutical agent for use in persons with kidney failure and some primary bone marrow disorders. Deficiency in an athlete would be unusual unless significant renal insufficiency was also present. Similarly, primary bone marrow disorders such as leukemia or multiple myeloma often cause significant anemia, but these disorders are rare in athletes.

Hemoglobin S in red blood cells protects against malaria and has been shown to reduce mortality from the disease compared with hemoglobin A.¹³ Sickle cell trait has long been believed to be a benign condition, but evidence is increasing that carriers experience an increased number of adverse events.¹⁴ Exertional sickling, which was first reported in military recruits more than 25 years ago, has been associated with sudden death during exercise.¹⁵ Causes of sudden death include metabolic acidosis, rhabdomyolysis, renal failure, and cardiac arrhythmia. Risk for adverse events is increased by conditions that can promote sickling of red blood cells, including intense exercise, particularly at a high altitude or in extreme heat. In a recent review of NCAA athletes who died suddenly, it was determined that the relative risk for sudden death is 37 times higher for an athlete with sickle cell trait.¹⁶

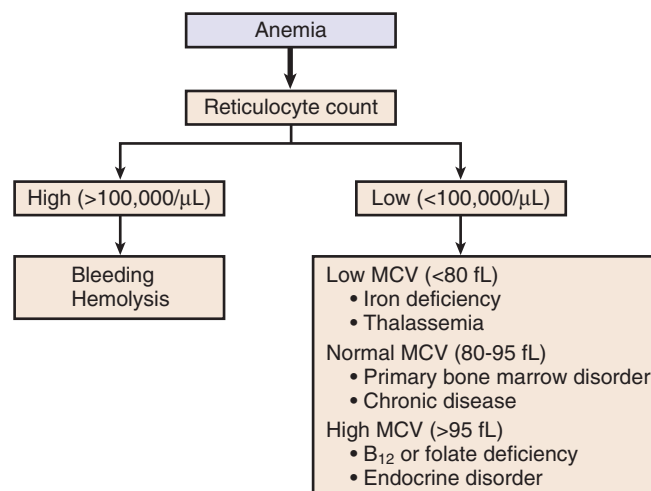


Fig. 16.1 An approach to the patient with anemia, based on reticulocyte count at presentation. MCV, Mean corpuscular volume.

Compared with athletes who have sickle cell trait, athletes with thalassemia have a much lower risk for sudden death. Athletes with either alpha or beta thalassemia minor often have a very mild anemia due to decreased production of hemoglobin A. Athletes with more significant disease will have significant anemia and may also have splenomegaly and skeletal abnormalities, which would likely preclude participation in competitive athletics.

Diagnosis

The diagnostic approach to the athlete with anemia is aimed at isolating the exact etiology in order to determine the appropriate therapy. A complete blood cell count (CBC) can be a useful guide to the initial evaluation. Significant abnormalities of white blood cells and platelets signal an occult, potentially serious bone marrow disorder that requires referral to a hematologist. However, an isolated anemia is much more common, and the workup can be guided by the CBC and reticulocyte count. Reticulocytes are immature red blood cells and are produced in higher numbers in the presence of bleeding or hemolysis. A high reticulocyte level indicates an appropriate bone marrow response to anemia and is not indicative of a bone marrow problem. Conversely, a low reticulocyte count is suggestive of anemia resulting from decreased red blood cell production (Fig. 16.1).

For disorders associated with decreased red blood cell production, evaluation of the size of red blood cells is pivotal in determining etiology. Microcytosis (MCV <80 fL) is most commonly associated with iron deficiency and thalassemia. Macrocytosis

(MCV >95 fL) may indicate deficiencies of either vitamin B₁₂ or folic acid as well as various endocrine or primary bone marrow disorders. Normocytic anemias (MCV 80–95 fL) can be induced by inflammation, bone marrow disorders, and early vitamin or mineral deficiency.

Iron deficiency is generally evaluated using a series of tests, including hemoglobin, hematocrit, MCV, serum iron, total iron-binding capacity, transferrin saturation, and serum ferritin. Although a bone marrow biopsy with Prussian blue staining for iron is generally regarded as the gold standard for diagnosing iron deficiency, serum ferritin is an excellent surrogate marker for iron stores in most patients. A serum ferritin level less than 10 to 15 ng/mL is nearly always suggestive of iron deficiency.¹⁷ However, higher ferritin values may also be suggestive of occult iron deficiency because a number of underlying inflammatory or malignant conditions can stimulate a threefold elevation in serum ferritin. The combination of low ferritin and decreased transferrin saturation is strongly suggestive of true iron deficiency.¹⁸

Screening for sickle cell disease is currently mandated in all 50 states and the District of Columbia.¹⁹ Screening programs use either isoelectric focusing or high-performance liquid chromatography, both of which have very high sensitivity and specificity. In 2010, the NCAA announced a sickle cell trait screening program for all Division 1 athletes.²⁰ This has been expanded to include all NCAA athletes despite concerns from many professional groups.²¹

A diagnosis of thalassemia minor can often be made by evaluating a CBC and family history. Thalassemia minor generally presents with severe microcytosis (MCV <70 fL) and mild or borderline anemia. Iron studies typically do not reveal iron deficiency, although some persons may have combined abnormalities. Hemoglobin electrophoresis is used to diagnose beta thalassemia, but results are normal in persons with alpha thalassemia, which is diagnosed by genetic testing.

For normocytic or macrocytic anemias, further testing should include assessment of levels of vitamin B₁₂, folate, and thyroid-stimulating hormone. Because of their larger size, a significant increase in reticulocytes can cause an increase in the MCV. Any athlete with an MCV greater than 110 fL should be referred to a hematologist.

Treatment

An adequate understanding of the various causes of anemia and related laboratory data is pivotal for adequate treatment of anemia in athletes (see Table 16.1). Athletes with iron deficiency anemia commonly present with the gradual onset of fatigue and decreased performance or exercise tolerance. A detailed history—including queries about gastrointestinal or urinary bleeding, menstruation, and nutritional practices—should be obtained. Characteristic laboratory findings include a microcytic hypochromic anemia; decreased serum ferritin, iron, and transferrin saturation levels; and elevated total iron-binding capacity. Additional testing to determine the etiology of iron deficiency (e.g., evaluation of the gastrointestinal tract) may be advised if necessary.

The replacement of iron in nonanemic iron-deficient athletes is controversial, although a recent review suggests that iron replacement improves the aerobic capacity of endurance athletes

with iron deficiency.²² One reasonable approach would be to first ensure that the athlete's diet included sufficient amounts of iron in the form of iron-rich foods such as lean meats, poultry, fish, grains, and vegetables. Dietary sources of heme iron have a higher bioavailability than do foods containing nonheme iron, and coingestion of foods containing ascorbic acid may enhance intestinal iron absorption. If dietary changes fail to improve the athlete's iron status, a therapeutic trial of oral iron supplementation may be useful.

Athletes with documented iron deficiency anemia should be treated with iron replacement therapy, typically with oral ferrous sulfate. Gastrointestinal adverse effects such as nausea or constipation may be dose limiting. Within 2 to 3 weeks, an increase in reticulocytes and MCV should be noted. Improvement in hemoglobin and ferritin levels may take several weeks. A minimum of 6 to 12 months of therapy is usually required, although some athletes may need more extended treatment depending on the underlying cause and severity of the anemia. Athletes who fail to respond to oral iron supplements or experience intolerable adverse effects with medication should be referred to a hematologist for consideration of alternative forms of iron replacement therapy.

The approach to athletes with sickle cell trait is controversial. The literature in this area largely consists of case reports or series; no prospective studies in athletes are available to guide the assessment of risk-reduction strategies. The American Heart Association and American College of Cardiology have issued a consensus statement on sickle cell trait and the athlete, which concludes that athletes should not be kept from participating, but preventative strategies and emergency response preparation are recommended.²³ Additionally, there are published guidelines for preventing sudden death in college and secondary school athletes that focus on risk reduction, gradual acclimatization to conditioning activities, and medical supervision.^{24,25} Because individuals with sickle cell trait may be more prone to exertional heat illness, recommendations for future research have focused on heat exposure, exercise intensity, and hydration as primary risk factors.²³

Return-to-Play Guidelines

Anemia generally does not preclude athletic participation. Athletes typically notice a decrement in athletic performance before they present with a severe anemia. For athletes with additional hematologic abnormalities, such as leukopenia or thrombocytopenia, referral to a hematologist is suggested before the athlete returns to full activity. For athletes with sickle cell trait, education and risk reduction are important components of safe participation in all athletics.¹⁸ Following these guidelines while closely monitoring the athlete for signs of difficulty should permit full participation.

DISORDERS OF HEMOSTASIS

Definition

Hemostasis is the process of blood clot formation at the site of an injured vessel. It is a complex interaction of platelets, von Willebrand factor (vWF), and a multitude of coagulation factors.²⁶

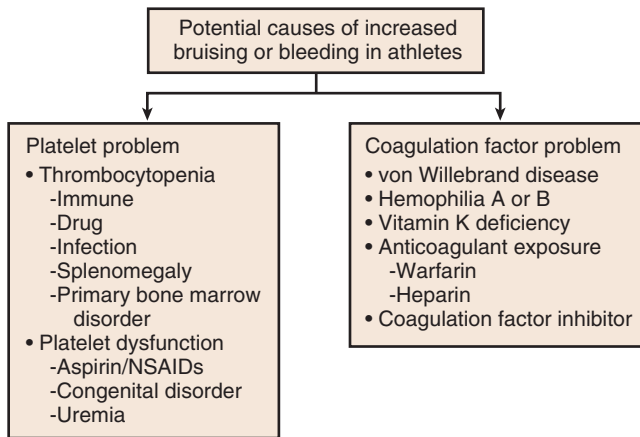


Fig. 16.2 Causes of increased bruising and bleeding in athletes. NSAIDs, Nonsteroidal antiinflammatory drugs.

Disorders of hemostasis can be acquired or congenital and can lead to spontaneous bleeding or increased bruising at the site of an injury (Fig. 16.2). Platelets are a critical part of the hemostatic plug that forms to stop bleeding acutely. Thrombocytopenia is defined by a platelet count of below 150,000/ μ L, and although it can be congenital, it is more often acquired. The risk of spontaneous bleeding does not increase significantly until the platelet count is less than 20,000/ μ L or the individual takes aspirin or nonsteroidal antiinflammatory drugs, both of which can inactivate platelets and enhance bleeding risk.

von Willebrand disease (vWD) is a congenital bleeding disorder that occurs as a result of reduced levels of vWF.²⁷ vWF has two important roles in hemostasis. First, it binds factor VIII in the blood, prolonging its half-life, and second, it forms a bridge between the endothelium of blood vessels and platelets to initiate hemostasis. Persons can bleed to varying degrees based on their levels of vWF. Persons with moderate or severe deficiency usually present with bleeding complications during childhood, whereas persons with mild deficiency may either have no symptoms or only mildly increased bleeding at mucosal sites.

Deficiencies of the many coagulation factors can occur and lead to increased bleeding. Two of the more common inherited deficiencies are hemophilia A and hemophilia B, which result from decreased production of factor VIII and factor IX, respectively.²⁸ As with vWD, moderate and severe deficiencies result in serious bleeding early in childhood, whereas persons with mild deficiencies may not have severe bleeding unless they undergo surgery or experience trauma.

Epidemiology

Disorders of hemostasis are uncommon among athletes, both because these disorders are relatively rare in the US population and because when they do occur, many are severe enough to prevent meaningful athletic participation. The incidence of idiopathic thrombocytopenic purpura (ITP) is 2 to 4 per 100,000 adults.²⁹ ITP can occur in both children and adults, and it is more common in women than in men. Drug-induced thrombocytopenia is also rare but can occur with exposure to many classes of drugs, including sulfa, antibiotics, and quinine.³⁰ The most common drug-induced thrombocytopenia is related to

heparin exposure, which would most likely occur either in the postoperative or perioperative setting.

vWD is the most common inherited bleeding disorder, affecting up to 1% of the population.²⁷ Persons with severe disease will generally present with bleeding in childhood, whereas milder disease may not be diagnosed until adulthood. Because the hemophilias are X-linked recessive traits, most cases present in males.²⁸ Hemophilia A occurs in about 1 in 5000 male births; hemophilia B is less common and presents in 1 in 30,000 male births. As with vWD, persons with severe disease present with bleeding very early in life, whereas persons with milder disease usually present much later in life due to a lack of spontaneous bleeding.³¹

Pathophysiology

The various etiologies of thrombocytopenia have different mechanisms. ITP is thought to be induced by autoantibodies that bind platelets, with subsequent removal from the circulation by the spleen.³² The antibodies cannot be detected clinically, and therefore the diagnosis of ITP is generally made by a process of exclusion. The antibodies can be induced by drug exposure or can be generated spontaneously, as with an autoimmune disorder. Drugs can also trigger decreased production of platelets, a mechanism that is dissimilar from the peripheral destruction of platelets observed in persons with ITP. Splenic enlargement (hypersplenism), in addition to other cytopenias, can also cause thrombocytopenia.³³ Chronic infections may also lead to thrombocytopenia, which is commonly noted in persons with human immunodeficiency virus and hepatitis C infection.^{34–36} In general, low platelets lead to increased bruising, mucosal bleeding, and, very rarely, spontaneous intracranial hemorrhage due to the loss of one of the primary mediators of hemostasis.

Bleeding from vWD is due to either decreased binding of platelets to damaged endothelium or the lack of vWF, which can cause decreased levels of factor VIII.²⁷ Bleeding is most often mucosal and typically presents as epistaxis or menorrhagia. The degree of bleeding is generally related to the levels of vWF, although other factors likely also contribute to a person's predisposition to bleeding.³⁷

Bleeding in persons with hemophilia is a result of decreased activity of either factor VIII or IX, each of which is critical for the integrity of the intrinsic clotting system. Patients with severe hemophilia (<1% of normal levels) have significant difficulty with spontaneous hemorrhage, most often in joints or muscles.¹⁷ Persons with milder deficiencies may not have spontaneous bleeding but will be at risk for significant hemorrhage with trauma or surgery.

Diagnosis

Initial evaluation of an athlete with increased bruising or bleeding should include a CBC, partial thromboplastin time (PTT), and prothrombin time (PT). The CBC will indicate the level of platelets, which may be extremely low in persons with ITP. The PTT is often prolonged in persons with vWD and is significantly prolonged in cases of hemophilia. To mitigate risk for significant bleeding, any athlete with significant abnormalities of platelets, PT, or PTT should be advised to refrain from all physical activity until a thorough evaluation has been conducted by a hematologist.

Isolated thrombocytopenia on a CBC may be induced by a variety of hematologic abnormalities. Occasionally platelets may clump as a result of anticoagulant in the blood sample, thus repeating a CBC is an appropriate first step.³² A history of other autoimmune diseases should increase suspicion for ITP. Because several drugs can cause thrombocytopenia, a detailed review of exposure to prescription drugs, over-the-counter medications, and supplement use is essential.^{29,30} Thrombocytopenia is also relatively common during pregnancy, which must be ruled out in a premenopausal woman. Athletes with more serious diseases, such as thrombotic thrombocytopenic purpura or leukemia, would likely have other symptoms and laboratory abnormalities, yet these diseases should be considered in the differential diagnosis.

Conditions that cause prolongations of the PT and PTT are also relatively common. Vitamin K deficiency and treatment with warfarin are common causes of prolonged PT, and vitamin K deficiency and heparin exposure can prolong PTT. Last, although the situation occurs very rarely, persons can acquire inhibitors to any of the coagulation factors, with factor VIII inhibitors being the most common.

Treatment

Conditions that affect the hemostatic system should be treated as directed after consultation with a hematologist. First-line treatment for ITP is systemic steroids—usually either prednisone or dexamethasone.²⁹ Most persons respond favorably with an increase in their platelet count, but many experience a relapse after discontinuing steroids, or a relapse may occur spontaneously several months later. If drug-induced ITP is suspected, the first step is to discontinue use of the drug and monitor platelet counts vigilantly. For persons who fail to respond to steroids or withdrawal of the offending drug, a bone marrow biopsy may be necessary to elucidate the etiology.

vWD can be treated with desmopressin, which stimulates release of vWD and factor VIII from storage pools in the body.²⁷ Patients undergo “desmopressin challenge,” and for those who respond, desmopressin is prescribed for use as needed for excessive bleeding. Potential adverse effects of desmopressin include hyponatremia, flushing, alterations in blood pressure, and headache.

The hemophilias are treated with replacement therapy. Both factors VIII and IX are available as recombinant factors that can be administered either for bleeding or prophylaxis against bleeding.^{38,39} Patients using replacement therapy should be followed up closely by a hematologist and should be evaluated emergently for either significant hemorrhage or lack of response to treatment at home.

Return-to-Play Guidelines

Decisions about returning to activity should be directed by the degree of hemostatic abnormality and the inherent “sport-specific”

risk. A runner who does not have a substantial risk for contact may return to activity with a platelet count below 50,000/ μ L. This approach is obviously not suitable for persons engaging in contact sports, and evaluation by a hematologist before clearance for participation is critical. It is also crucial that platelet levels be monitored closely, because conditions such as ITP are prone to relapse. A similar algorithm should be adhered to for athletes with either vWD or hemophilia. The degree of deficiency, characteristics of any recent injury, and the amount of contact in the athlete’s sport are all vital considerations that facilitate informed medical decisions about return to play by the hematologist and sports medicine staff.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Naik RP, Haywood C. Sick Cell Trait diagnosis: clinical and social implications. *Hematology Am Soc Hematol Educ Program*. 2015;2015:160–167.

Level of Evidence:

VII

Summary:

This review provides an excellent overview of the current state of knowledge of sickle cell trait, implications of testing and the areas of greatest need for research.

Citation:

Burden RJ, Morton K, Richards T, et al. Is iron treatment beneficial in, iron-deficient but non-anemic (IDNA) endurance athletes? A systematic review and meta-analysis. *Br J Sports Med*. 2015;49:1389–1397.

Level of Evidence:

I

Summary:

This article provides a thorough review of potential benefits of treating of iron deficiency without anemia in athletes.

Citation:

Maron BJ, Harris KM, Thompson PD, et al. Eligibility and disqualification recommendations for competitive athletes with cardiovascular abnormalities: Task Force 14: Sick Cell Trait: A scientific statement from the American Heart Association and American College of Cardiology. *J Am Coll Cardiol*. 2015;66:2444–2446.

Level of Evidence:

VII

Summary:

This report from a task force from the American College of Cardiology and American Heart Association provides recommendations for athletes with sickle cell trait.

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